

Causal Inference in Economics and Marketing

Section 08: Heterogeneous Treatment Effects and Policy Learning

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1 From “does it work” to “for whom” and “whom to treat”

1.1 Three questions

Sections 02 to 07 answered one question: what is the average effect of a treatment? The average treatment effect (ATE), the ATT, and the ATT(g,t) of the last section are all averages. They inform a uniform decision: roll the policy out to everyone, or to no one.

Two further questions sit behind most applied work.

- **For whom does it work?** The conditional average treatment effect (CATE), $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, describes how the effect varies with observed characteristics.
- **Whom should we treat?** A policy $\pi(x) \in \{0, 1\}$ assigns treatment from characteristics. Its value is the average gain it produces.

A positive ATE can hide a subgroup that is harmed. A null ATE can hide a subgroup that responds strongly. Marketing lives on this: a discount that is worthless on average may pay for itself among price-sensitive customers. Development economics lives on it too: a cash transfer may be worth most to households that are not the poorest.

1.2 Why the CATE is a harder object

Every target so far was a number. The CATE is a function. That changes what we can hope for.

- Estimating a function is a prediction problem. Prediction rates are slower than $n^{-1/2}$ and confidence intervals do not come for free.
- Even with both potential outcomes observed, estimating $\tau(x)$ would be as hard as estimating the regression of $Y(1) - Y(0)$ on X .
- Genovese and Wasserman (2008) show that adaptive confidence bands for a curve do not exist in general. Bands must target a simpler surrogate, or restrict the dimension of X .

The response is to relax the goal, and the lecture is organized by how much we relax it (Chernozhukov et al. 2026, ch. 14).

Goal	What we get	Section
A low-dimensional summary: best linear predictor or group effects	Root- n inference, uniform bands	4
Confidence intervals at a point with forests	Valid in low dimensions, brittle otherwise	4.5
A flexible CATE model with small mean squared error	No intervals; out-of-sample scores and tests	3, 5
Tests that a model has found heterogeneity	Calibration, targeting curves	6
A good policy, without caring about the CATE's magnitude	Regret bounds, honest values	7

1.3 Identification

Conditional exogeneity, $D \perp (Y(0), Y(1)) \mid Z$, identifies the CATE for any $X \subseteq Z$:

$$\tau(X) = E[g(1, Z) - g(0, Z) \mid X], \quad g(d, z) = E[Y \mid D = d, Z = z].$$

Two remarks matter later.

- X may be a small subset of the controls Z . Confounding is handled by Z ; heterogeneity is described by X . The high-dimensional CATE $\delta(Z) = E[Y(1) - Y(0) \mid Z]$ and its coarsening $\tau(X) = E[\delta(Z) \mid X]$ are different objects.
- On Z , the effect among the treated equals the effect among the controls: $E[Y(1) - Y(0) \mid Z, D = 1] = E[Y(1) - Y(0) \mid Z, D = 0] = \delta(Z)$. This fails for a subset X , because the distribution of Z given X differs across arms. The X-learner of Section 3.3 is built on the first fact.

1.4 The first CATE model: an interaction regression

The simplest CATE model is the regression with interactions, $Y = \beta_0 + \beta_1 D + \beta_2' X + D X' \beta_3 + e$, which gives $\hat{\tau}(x) = \hat{\beta}_1 + x' \hat{\beta}_3$ (Facure 2023, ch. 18). We use a simulated design throughout. Think of D as a marketing nudge, Y as spending, and $x_1, \dots, x_5$ as customer characteristics. The true CATE is $1 + x_1 + 0.5(x_2^2 - 1)$; assignment depends on x_1 and x_2 ; the baseline is nonlinear in x_2 .

```
dat <- sim_hte(n = 4000, dgp = "smooth", seed = 1)
train <- dat[1:2000, ]
score_set <- dat[2001:3000, ]
test <- dat[3001:4000, ]
fit_int <- lm(y ~ d * (x1 + x2 + x3 + x4 + x5), data = train)
tau_int <- coef(fit_int)["d"] + as.matrix(test[, x5]) %*% coef(fit_int)[paste0("d:", x5)]
c(rmse = sqrt(mean((tau_int - test$tau_true)^2)), cor = cor(tau_int, test$tau_true))
#>      rmse      cor
#> 0.9853457 0.6924574
```

The regression catches the linear part of the heterogeneity in x_1 and misses the curvature in x_2 . It also has no protection against misspecification of the baseline: the terms in X must absorb confounding and describe heterogeneity at the same time. Figure 1 shows both facts.

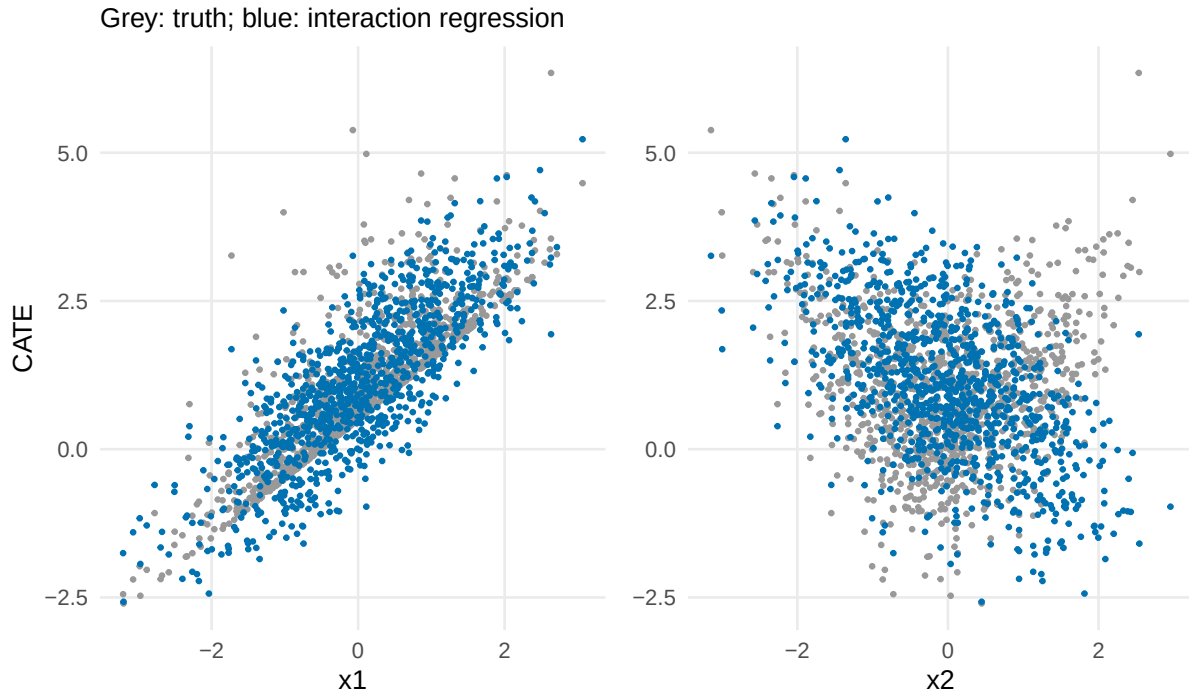


Figure 1: Interaction-regression CATE against the truth on the test sample. Left: along x_1 the linear model tracks the truth. Right: along x_2 it cannot bend.

The rest of the lecture separates the two jobs. Confounding is removed once, by cross-fitted nuisances. Heterogeneity is then a regression of a single constructed variable on X .

1.5 Roadmap

Section 2 builds that variable, the doubly robust pseudo-outcome, and shows why its conditional mean is the CATE. Section 3 turns any regression method into a CATE model and compares the meta-learners in three designs. Section 4 gives valid inference on summaries and on forests. Sections 5 and 6 select and validate a model out of sample. Section 7 learns and evaluates policies. Section 8 collects traps, Section 9 the replications and a checklist. The appendix holds the formal statements.

2 One signal, many uses: the doubly robust pseudo-outcome

2.1 Definition and the key property

Let $p(z) = P(D = 1 \mid Z = z)$ and $g(d, z) = E[Y \mid D = d, Z = z]$, with $\eta = (p, g)$. The doubly robust pseudo-outcome is

$$Y(\eta) = g(1, Z) - g(0, Z) + H(p) (Y - g(D, Z)), \quad H(p) = \frac{D}{p(Z)} - \frac{1 - D}{1 - p(Z)}.$$

It is the summand of the AIPW estimator of Section 04. Its mean is the ATE. Its conditional mean is the CATE:

$$E[Y(\eta_0) \mid X] = \tau(X).$$

Skeleton. Condition on Z first. Given Z , $E[H(p_0)(Y - g_0(D, Z)) \mid Z] = 0$ because $E[D(Y - g_0(1, Z)) \mid Z] = p_0(Z) \cdot 0$ and likewise for the control term. What remains is $g_0(1, Z) - g_0(0, Z) = \delta(Z)$. Then average over Z given X .

The property that makes everything else work is **conditional Neyman orthogonality**: the derivative of $E[Y(\eta) \mid X]$ with respect to η at η_0 is zero. Errors in $\hat{p}$ and $\hat{g}$ enter the conditional mean only through their product,

$$E[Y(\eta_0) - Y(\hat{\eta}) \mid X] = E[(H(p_0) - H(\hat{p}))(g_0(D, Z) - \hat{g}(D, Z)) \mid X],$$

so the pseudo-outcome is as good as the true one whenever the product of the two nuisance errors is small. Appendix 10.1 proves the identity in three lines.

2.2 Cross-fitting the nuisances

As in Section 04, the nuisances are cross-fitted: $\hat{p}$ and $\hat{g}$ are trained on folds that exclude the observation they predict. The final regression of $Y(\hat{\eta})$ on X may then use all rows (Chernozhukov et al. 2026, Remark 15.1.1). Overfitting in the first stage cannot leak into the second, because the label of row i never saw row i .

2.3 Code: building the scores

`dr_scores()` builds the vector once. Every later function consumes the returned object.

```
sc_train <- dr_scores(train, y = "y", d = "d", x = x5, seed = 1)
sc_train
#> Doubly robust pseudo-outcomes (dr)
#>   n = 2000, treated share = 0.494, folds = 5
#>   nuisances: p = classif.log_reg, mu0 = regr.lm, mu1 = regr.lm
#>   ATE (mean of the score) = 1.1333 (SE 0.1101)
#>   score quantiles (1%, 50%, 99%): -8.761,  0.783, 18.167
```

With linear nuisances the outcome model is misspecified (the baseline is quadratic in x_2) while the logistic propensity model is correctly specified. Random forests fit both nuisances flexibly. In a simulation we can also plug in the true nuisances, which no analyst has. The regression of the score on the true CATE, whose slope should be one, summarizes how close each version gets.

```
rf_first <- hte_forest(min_node = 20, trees = 300)
rf_final <- hte_forest(min_node = 100, trees = 300)
sc_train_rf <- dr_scores(train, "y", "d", x5, learner_mu = rf_first, seed = 1)
sc_train_oracle <- dr_scores(train, "y", "d", p_hat = "p_true", mu0_hat = "mu0_true", mu1_hat = "mu1_true")
slope_tab <- sapply(list(linear = sc_train, forest = sc_train_rf, true_nuisances = sc_train_oracle), function(s) {
  b <- cate_blp(s, ~ tau_true, uniform = FALSE)$coefficients
  c(slope_on_true_cate = b$estimate[2], hci_se = b$std.error[2], sd_score = sd(s$score))
})
round(slope_tab, 3)
#>           linear forest true_nuisances
#> slope_on_true_cate 1.378  1.197      0.963
#> hci_se             0.133  0.083      0.044
#> sd_score           4.924  3.136      2.534
```

The slope moves toward one and the score's standard deviation falls as the nuisances improve. The gap that remains with estimated nuisances is the product term above: small on average, but concentrated where the outcome error and the propensity error are both large, which here is the tail of x_2 . Figure 2 checks the key property with the forest nuisances, which the rest of the lecture uses: a running mean of the score along x_1 follows the truth.

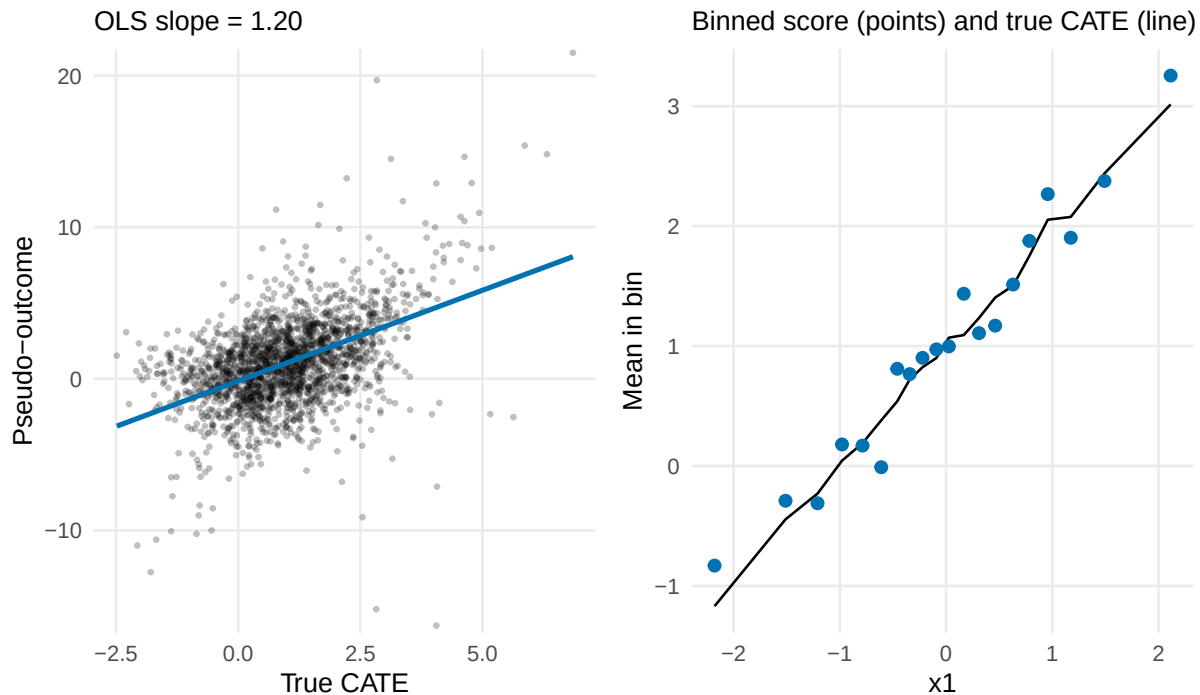


Figure 2: The conditional mean of the pseudo-outcome is the CATE. Left: the score against the true CATE, with the OLS line (slope reported). Right: a binned mean of the score along x_1 (points) against the true CATE averaged in the same bins (line).

The score is noisy: an individual value divides a residual by a propensity. That noise is the price of unbiasedness at the individual level, and it is why the rest of the lecture never looks at one score alone. It averages them, regresses them, or classifies their sign.

2.4 Two alternatives and why they lose

`dr_scores(type = "ipw")` returns $YH(\hat{p})$, the inverse-propensity signal, whose conditional mean is also the CATE. `type = "reg"` returns $\hat{g}(1, Z) - \hat{g}(0, Z)$, the plug-in signal. The first is unbiased only if the propensity is right and has much larger variance; the second is unbiased only if the outcome model is right and has no variance protection. The doubly robust signal combines them.

```
sc_ipw <- dr_scores(train, "y", "d", x5, type = "ipw", seed = 1)
sc_reg <- dr_scores(train, "y", "d", x5, type = "reg", learner_mu = rf_first, seed = 1)
type_tab <- supply(list(dr = sc_train_rf, ipw = sc_ipw, reg = sc_reg), function(s) {
  b <- cate_blp(s, ~ tau_true, uniform = FALSE)$coefficients
  c(slope_on_true_cate = b$estimate[2], hci_se = b$std.error[2], sd_score = sd(s$score))
})
round(type_tab, 3)
#>           dr    ipw    reg
#> slope_on_true_cate 1.197 1.449 0.868
#> hci_se             0.083 0.182 0.017
#> sd_score           3.136 6.309 1.221
```

The plug-in signal has the smallest variance and a slope below one: a regularized outcome model shrinks the heterogeneity it cannot represent. The inverse-propensity signal needs only the propensity, which is right here, but its standard deviation is 2.0 times that of the doubly robust signal. The doubly robust signal keeps the slope of the inverse-propensity signal at a fraction of its variance.

2.5 Connection to Section 04

The mean of the score is the AIPW estimate of the ATE, with the same standard error as `est_aipw()`:

```
tidy(sc_train_rf)
#>   term estimate std.error statistic      p.value  conf.low conf.high
#> 1  ATE 1.013221 0.07012929  14.4479 2.584403e-47 0.8757702  1.150672
```

Everything below is a regression of the score on something: on a dictionary of covariate functions (Section 4), on a machine learner (Section 3), on a candidate model’s own predictions (Section 6), or on the sign of a policy (Section 7).

3 Meta-learners: turning a regression method into a CATE model

3.1 The idea and a notation

A meta-learner decomposes CATE estimation into regression problems and hands each to an off-the-shelf method (Künzel et al. 2019). Write $O_H(\{X_i, L_i, W_i\})$ for a regression oracle: it takes covariates, labels, and optional weights and returns an estimate of the weighted best predictor $\arg\min_{h \in H} E[W(L - h(X))^2]$ (Chernozhukov et al. 2026, Section 15.1). Every meta-learner below ends with one call to this oracle on the heterogeneity variables X . The learners differ in the label L_i they construct, and in how they construct it. All first stages are cross-fitted; the final call uses all rows.

3.2 S- and T-learners

The **S-learner** fits one model of Y on (D, Z) and sets $L_i = \hat{g}(1, Z_i) - \hat{g}(0, Z_i)$. The **T-learner** fits one model per arm, $\hat{g}_1$ on the treated and $\hat{g}_0$ on the controls, and sets $L_i = \hat{g}_1(Z_i) - \hat{g}_0(Z_i)$. Both converge to the best approximation of the CATE in the final oracle’s class if the outcome models are consistent. Both fail in characteristic ways.

- The S-learner regularizes the treatment like any other covariate. When the effect is small relative to the baseline variation, a penalized or tree-based learner shrinks it toward zero, or drops D entirely.
- The T-learner cannot share information across arms. When one arm is small, its model is heavily regularized while the other is flexible, and the difference of two mismatched fits produces spurious heterogeneity.

The second failure is Figure 3, on the first design of Chernozhukov et al. (2026, Example 15.1.1): 5 percent treated, a constant effect of 0.5, a baseline with a jump on $[0.6, 0.8]$, and noise of 0.05.

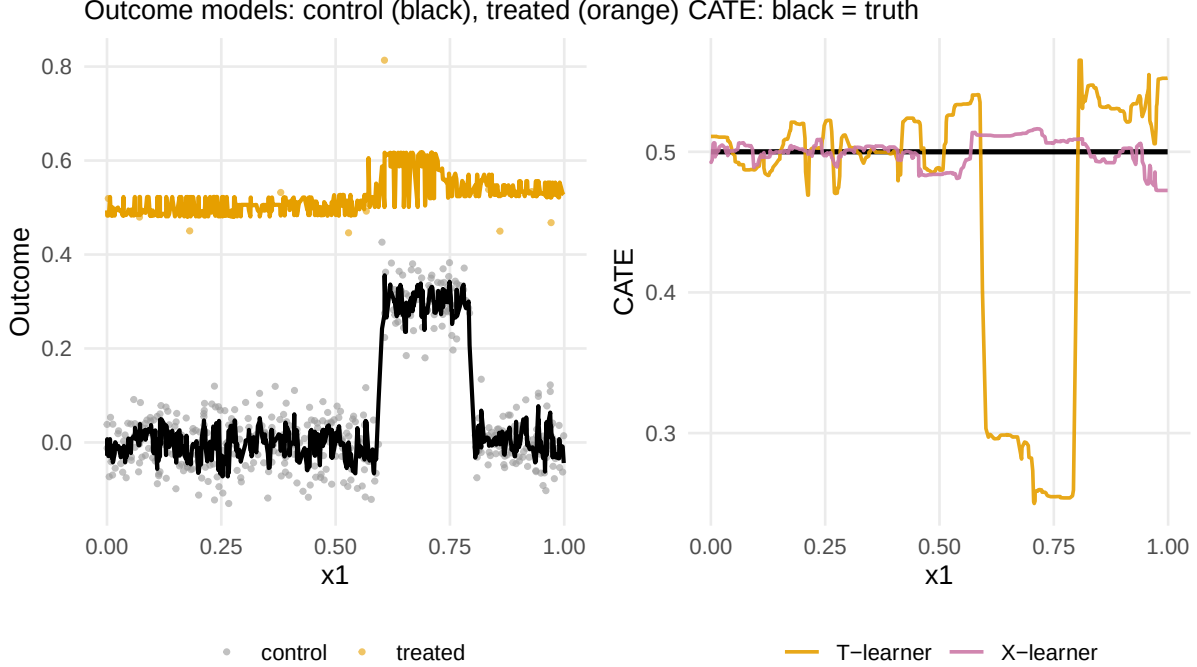


Figure 3: Design 1 (5 percent treated, constant effect, jump in the baseline), one sample of $n = 500$. Left: the cross-fitted outcome models of the T-learner; the control model finds the jump, the treated model, fitted on about 25 points, finds it only noisily. Right: the resulting CATE estimates of the T- and X-learners against the truth.

3.3 X-learner

The X-learner (Künzel et al. 2019) uses the fact that the effect is the same among treated and controls once we condition on Z . Each arm can therefore estimate the CATE with only *one* imputed counterfactual:

$$\delta^T(Z) = E[Y - g_0(Z) \mid Z, D = 1], \quad \delta^C(Z) = E[g_1(Z) - Y \mid Z, D = 0].$$

The treated version needs only the control model g_0 , which is estimated on the large arm; the control version needs only g_1 . The two estimates are blended with the propensity, $\hat{\delta}(Z) = (1 - \hat{p})\hat{\delta}^T + \hat{p}\hat{\delta}^C$, so that where treatment is rare, the version that uses the well-estimated control model gets the weight. The label is $L_i = \hat{\delta}(Z_i)$. The right panel of Figure 3 shows the repair.

Two refinements come from the covariate-shift view: $\hat{\delta}^T$ is trained on the treated but evaluated everywhere, so training it with weights $(1 - \hat{p})^2/\hat{p}$ (and $\hat{\delta}^C$ with $\hat{p}^2/(1 - \hat{p})$) targets the population where it will be used (`adapt = TRUE`, the domain-adapted X-learner). And the X-learner's outcome models can feed the DR-learner below in place of the T-learner's.

3.4 DR-learner

The **DR-learner** regresses the pseudo-outcome of Section 2 on X : $L_i = Y_i(\hat{\eta})$ (Kennedy 2023; Foster and Syrgkanis 2023). Its error bound is

$$E[(\hat{\tau} - \tau^*)^2] \lesssim r_n^2 + \text{Err}(\hat{g})^2 \text{Err}(H(\hat{p}))^2,$$

where r_n^2 is the rate the oracle would achieve with the true label and τ^* is the best approximation in the oracle's class. Only the product of the nuisance errors enters: this is double robustness for a function (Appendix 10.2 sketches the argument). The price is variance. The label divides a residual by $\hat{p}$ or $1 - \hat{p}$, and when overlap is weak in parts of the covariate space the label takes extreme values there.

3.5 R-learner

The **R-learner** (Nie and Wager 2021) starts from Robinson’s decomposition. If the CATE depends on X only, then

$$Y - E[Y \mid Z] = \tau(X) (D - E[D \mid Z]) + \epsilon, \quad E[\epsilon \mid D, Z] = 0,$$

so τ minimizes $E[(\tilde{Y} - \tau(X)\tilde{D})^2]$ with $\tilde{Y} = Y - l(Z)$, $\tilde{D} = D - p(Z)$. Rewriting the loss as $E[\tilde{D}^2(\tilde{Y}/\tilde{D} - \tau(X))^2]$ makes it a weighted regression: label $L_i = \tilde{Y}_i/\tilde{D}_i$, weight $W_i = \tilde{D}_i^2$. The final oracle must accept weights. The error bound is

$$E[(\hat{\tau} - \tau_0)^2] \lesssim r_n^2 + \text{Err}(\hat{p})^4 + \text{Err}(\hat{p})^2 \text{Err}(\hat{l})^2,$$

so the propensity matters more than in the DR-learner, but nothing is divided by it.

What does the R-learner estimate when the CATE depends on more than X , or when the oracle’s class is misspecified? Not the best approximation of τ_0 , but an overlap-weighted one:

$$E[(\tilde{Y} - \tau(X)\tilde{D})^2] = E[(\delta(Z) - \tau(X))^2 \text{Var}(D \mid Z)] + E[\epsilon^2].$$

Skeleton. Substitute $\tilde{Y} = \delta(Z)\tilde{D} + \epsilon$, expand, use $E[\epsilon \mid D, Z] = 0$ to kill the cross term, and $E[\tilde{D}^2 \mid Z] = \text{Var}(D \mid Z)$. Regions where treatment is nearly deterministic get no weight. That is a bias relative to the DR-learner’s target, and a large variance reduction.

3.6 Guidelines and an experiment

Table 2 condenses the discussion.

Table 2: Meta-learners at a glance.

Learner	Needs	Converges to	Weak when
S	outcome model	best approximation of the CATE	effect small relative to baseline (shrinks it)
T	outcome models	best approximation	one arm is small
X	outcome models (+ propensity to blend)	best approximation	CATE as complex as the baseline
DR	outcome or propensity model	best approximation	overlap weak (extreme labels)
R	propensity and outcome model	overlap-weighted approximation	propensity poorly estimated; ignores low-overlap regions

Two experiments make the guidelines concrete. The first uses the three designs of Chernozhukov et al. (2026, Example 15.1.1) with random-forest oracles: design 1 as above; design 2 with a simple baseline and the jump in the CATE, 5 percent treated; design 3 like design 2 with 95 percent treated. Learners are fitted on 70 percent of a sample, the two ensembles of Section 5.4 are stacked on the remaining 30 percent, and the root mean squared error is computed against the true CATE on a fresh sample.

Three patterns stand out.

- Design 1 rewards outcome modelling with borrowed strength: the X-learner is best (X-learner at $n = 500$), because the control model is easy and the effect is constant. The learners that divide by the propensity pay for it: with about 17 treated units in the training part, the DR-learner’s label multiplies their residuals by 20.
- Design 3 rewards the T-learner: with 95 percent treated the treated model is the good one and the effect is complex, so borrowing the control model hurts (T-learner is best at $n = 500$).
- Growing n from 500 to 2000 lowers every learner’s error but does not change which learner is best within a design. The learners that divide by the propensity remain behind with forest final stages even at $n = 2000$: their labels are noisy, and a forest fitted to noisy labels needs large leaves and many

Table 3: Root mean squared error of the CATE (mean and standard deviation over 60 simulations) with random-forest oracles. Design 1: constant effect, jump in the baseline, 5 percent treated. Design 2: jump in the effect, simple baseline, 5 percent treated. Design 3: as design 2 with 95 percent treated.

Design	n	S	T	X	DA X	DR	R	Q-agg.	Best
DGP 1	500	0.056 (0.032)	0.080 (0.027)	0.025 (0.008)	0.026 (0.009)	0.167 (0.099)	0.102 (0.052)	0.077 (0.045)	0.077 (0.046)
	2000	0.019 (0.008)	0.034 (0.005)	0.028 (0.003)	0.028 (0.004)	0.094 (0.031)	0.074 (0.013)	0.026 (0.011)	0.026 (0.012)
DGP 2	500	0.166 (0.019)	0.143 (0.032)	0.143 (0.031)	0.141 (0.027)	0.232 (0.139)	0.148 (0.067)	0.172 (0.067)	0.174 (0.066)
	2000	0.123 (0.010)	0.062 (0.013)	0.063 (0.013)	0.063 (0.013)	0.132 (0.049)	0.092 (0.017)	0.070 (0.022)	0.072 (0.024)
DGP 3	500	0.110 (0.011)	0.045 (0.010)	0.131 (0.025)	0.133 (0.024)	0.069 (0.028)	0.123 (0.051)	0.056 (0.023)	0.057 (0.026)
	2000	0.091 (0.005)	0.031 (0.005)	0.060 (0.011)	0.061 (0.011)	0.071 (0.013)	0.082 (0.017)	0.033 (0.005)	0.033 (0.006)

observations. The ensembles track the best base learner in every design without being told which one it is.

The book obtains the same qualitative picture with gradient-boosted oracles, with the DR- and R-learners closer to the front in designs 1 and 2. The lesson is that the ranking depends on the oracle, the sample size, and the design: there is no single best meta-learner, and Section 5 replaces the guesswork with a data-driven choice.

The second experiment isolates double robustness. It uses the running “smooth” design with *linear* oracles. The outcome regressions are misspecified (the baseline is quadratic in x_2) while the logistic propensity model is exactly right. The final stage is linear in x_1 and x_2^2 , which contains the true CATE.

Table 4: Misspecified linear outcome models, correct logistic propensity; $n = 2000$, 200 simulations. Bias is the root mean square over test points of the mean error across simulations; the standard deviation is the root mean square of the simulation standard deviation; RMSE combines both.

Learner	Bias	Standard deviation	RMSE
S-learner	1.233	0.100	1.237
T-learner	0.713	0.152	0.729
X-learner	0.711	0.152	0.727
DA X-learner	0.617	0.156	0.636
DR-learner	0.022	0.595	0.595
R-learner	0.024	0.312	0.313

The S-, T-, and X-learners inherit the outcome model’s bias and cannot remove it with more data; their errors are almost entirely bias. The DR- and R-learners have almost no bias, because the correct propensity model protects them as the error bounds promise, and their errors are variance, which shrinks with n . This is the reason to prefer them whenever the outcome model is the shaky part of the analysis, which in observational data it usually is.

3.7 Multi-arm and other extensions

With K arms the S- and T-learners extend directly (one model with the arm as a feature, or K models), and pairwise X- and R-learners can be built for each contrast. The R-learner then depends on the reference arm, and Zhou, Zhang, and Tu (2023) propose a reference-free version that estimates $E[Y^{(k)} - Y \mid X]$ for every arm under a simplex constraint. Neural networks with a shared representation and a balancing penalty (Shalit, Johansson, and Sontag 2017) are another route to outcome modelling under covariate shift. None of these change the logic of this section: a pseudo-label, then a regression.

3.8 Interpreting a black-box CATE

A forest-based $\hat{\tau}$ is not readable. Two devices restore interpretation without new assumptions.

- **Distillation:** fit a shallow regression tree to $\hat{\tau}(X_i)$ on X_i and read its splits.
- **Group differences:** compare covariate means between the top and bottom groups of $\hat{\tau}$ (Section 6 reports them).

```
dr_fit <- cate_learner(scores = sc_train_rf, x_het = x5, method = "dr", learner_final = rf_final)
tau_test <- predict(dr_fit, test)
distill <- rpart::rpart(tau_hat ~ ., data = data.frame(tau_hat = tau_test, test[, x5]),
  control = rpart::rpart.control(maxdepth = 2, minbucket = 50, cp = 0.01))

distill
#> n= 1000
#>
#> node), split, n, deviance, yval
#>      * denotes terminal node
#>
#> 1) root 1000 2213.81200 1.0336880
#>   2) x1< 0.07639589 527 724.65080 0.2495114
#>     4) x2< 1.346754 477 378.75430 0.1009114 *
#>     5) x2>=1.346754 50 234.87770 1.6671550 *
#>   3) x1>=0.07639589 473 804.02410 1.9073910
#>     6) x1< 1.715707 423 632.41140 1.7526680 *
#>     7) x1>=1.715707 50 75.81833 3.2163460 *
```

The distillation is fitted to predictions on the test sample, never to in-sample fitted values of a forest, which overfit their own noisy labels. Its root splits on x_1 , the variable that drives the effect linearly in the design; the splits below use x_2 .

4 Inference on summaries of the CATE

4.1 Best linear predictor

The most useful relaxation replaces $\tau(x)$ by its best linear approximation in a dictionary $p(x)$ of $d \ll n$ functions:

$$\beta_0 = \arg \min_{\beta} E[(\tau(X) - p(X)' \beta)^2] = E[p(X)p(X)']^{-1} E[p(X) \tau(X)] = E[p(X)p(X)']^{-1} E[p(X) Y(\eta_0)],$$

where the last equality uses $E[Y(\eta_0) | X] = \tau(X)$. So β_0 is the population regression coefficient of the pseudo-outcome on $p(X)$, and its sample analogue is ordinary least squares of $Y_i(\hat{\eta})$ on $p(X_i)$ (Semenova and Chernozhukov 2021; Chernozhukov et al. 2026, Section 14.2).

Skeleton of why OLS inference is valid. Write $\hat{\beta} - \beta_0 = \hat{Q}^{-1} \mathbb{E}_n[p(X)(Y(\hat{\eta}) - p(X)' \beta_0)]$ with $\hat{Q} = \mathbb{E}_n[p(X)p(X)']$. Replace $Y(\hat{\eta})$ by $Y(\eta_0)$: the difference has conditional mean equal to the product of the nuisance errors (Section 2), which is $o(n^{-1/2})$ under the rate condition of Section 04, and a conditional variance that vanishes with consistency of the nuisances. What remains is OLS with the oracle outcome, whose HC sandwich is the right variance. The dictionary can be group indicators, polynomials, or splines. When it is group indicators the coefficients are group average treatment effects (GATEs).

`cate_blp()` runs the regression, reports HC1 standard errors, and adds a simultaneous band from the multiplier bootstrap of Section 07 applied to the influence functions $\hat{Q}^{-1} p(X_i) \hat{e}_i$.

```
blp <- cate_blp(sc_train_rf, ~ x1 + I(x2^2), seed = 1)
blp
#> Best linear predictor of the CATE: score ~ x1 + I(x2^2)
#> n = 2000, 95% pointwise |z| = 1.96, simultaneous = 2.336
#>      term estimate std.error conf.low conf.high band.low band.high
#> (Intercept)  0.2082  0.09066  0.03054  0.3859 -0.003584  0.4201
#>      x1      0.9994  0.07040  0.86145  1.1374  0.834961  1.1639
#>      I(x2^2)  0.8461  0.09905  0.65197  1.0402  0.614697  1.0775
#> Wald test of no heterogeneity: chi2(2) = 255.92, p = 2.67e-56
blp_oracle <- cate_blp(sc_train_oracle, ~ x1 + I(x2^2), uniform = FALSE)
blp_oracle$coefficients %>% select(term, estimate, std.error) %>%
  mutate(across(where(is.numeric), ~ round(., 3)))
#>      term estimate std.error
```

```
#> 1 (Intercept) 0.423 0.060
#> 2          x1 0.942 0.051
#> 3      I(x2^2) 0.508 0.038
```

The true CATE is $0.5 + x_1 + 0.5x_2^2$, so the three coefficients target 0.5, 1, and 0.5. The coefficient on x_1 is on target. The coefficient on x_2^2 is 0.85 with the forest nuisances and 0.51 with the true nuisances: the difference is the product-of-errors term of Section 2, which is largest in the tail of x_2 , where the forest's outcome error and the propensity's leverage are both large. Better nuisances, not a different second stage, close that gap. The Wald test rejects a constant effect. The simultaneous critical value is above 1.96 because it covers all three coefficients at once.

Figure 4 evaluates the curve along x_1 with pointwise and simultaneous bands. The simultaneous band covers the whole curve with probability 0.95; the pointwise band covers one pre-specified point.

```
grid <- data.frame(x1 = seq(-2.5, 2.5, by = 0.1), x2 = 0)
plot_cate_blp(bl, grid, seed = 1) +
  geom_line(data = grid, aes(x1, 0.5 + x1), linetype = "dashed", colour = "black")
```

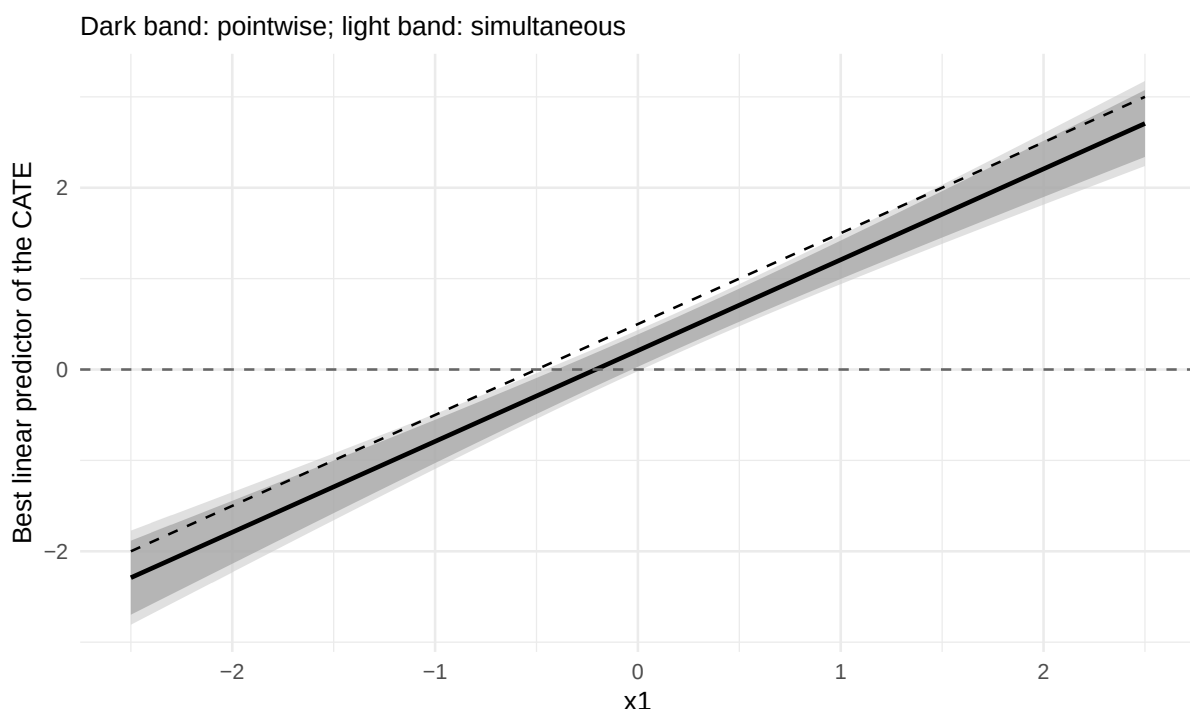


Figure 4: Best linear predictor of the CATE along x_1 (x_2 fixed at 0) with pointwise (dark) and simultaneous (light) 95 percent bands. The dashed line is the true CATE.

4.2 Group average treatment effects

With group indicators as the dictionary, `cate_gate()` returns the doubly robust GATEs with pointwise and simultaneous intervals and a joint test of equality.

```
q_x1 <- cut(train$x1, quantile(train$x1, 0:4 / 4), include.lowest = TRUE, labels = paste0("Q", 1:4))
gate <- cate_gate(sc_train_rf, q_x1, seed = 1)
gate
#> Group average treatment effects (doubly robust)
#>   group  n share estimate std.error conf.low conf.high band.low band.high
#>   Q1 500 0.25 -0.2162    0.1366  -0.4839  0.05146  -0.5590  0.1266
#>   Q2 500 0.25  0.6882    0.1402   0.4134  0.96301  0.3363  1.0401
#>   Q3 500 0.25  1.2449    0.1267   0.9965  1.49334  0.9269  1.5630
#>   Q4 500 0.25  2.3360    0.1326   2.0761  2.59584  2.0032  2.6687
```

```
#> Wald test of equal effects: chi2(3) = 189.41, p = 8.17e-41
```

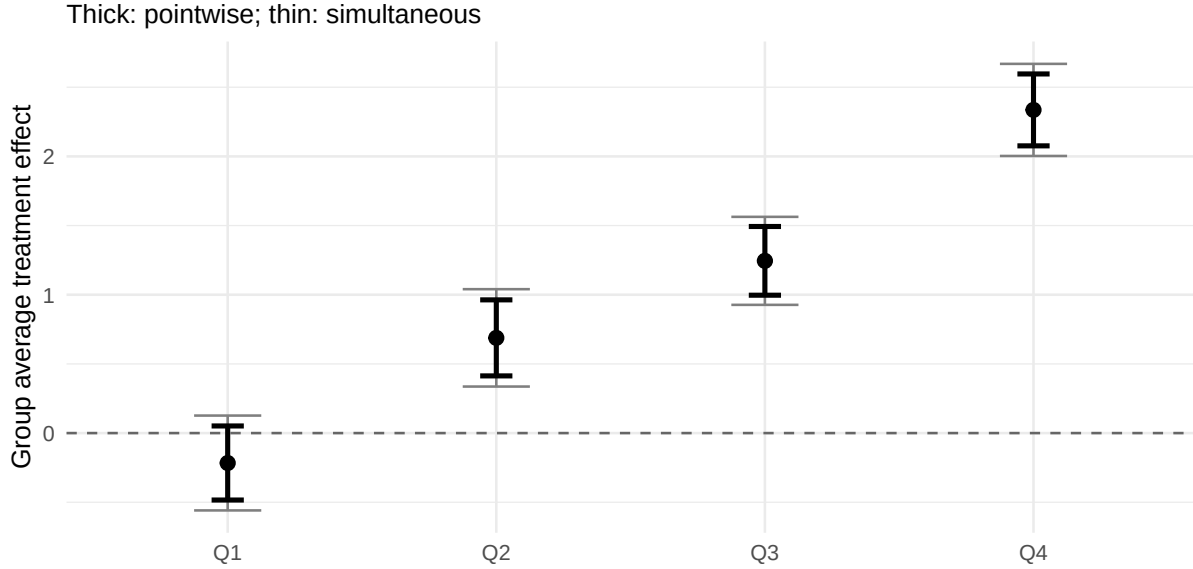


Figure 5: Group average treatment effects by quartile of x_1 . Thick bars: pointwise 95 percent intervals; thin bars: simultaneous intervals valid for all four groups at once.

The choice between pointwise and simultaneous intervals is a choice of question. A pre-registered interest in the bottom quartile alone uses the pointwise interval. A claim that “effects differ across quartiles” needs the joint test or the simultaneous intervals.

4.3 A subset of covariates is enough

The dictionary used x_1 and x_2 while confounding was handled by all five covariates inside the scores. Nothing required $X = Z$. That is the practical advantage of the pseudo-outcome route: heterogeneity can be described in the two or three variables a decision maker can act on, while the nuisances use everything.

4.4 Continuous treatments

With a continuous treatment the natural target is the derivative $E[\partial_t E[Y | T = t, X] | X]$ (Facure 2023, ch. 18). Under the partially linear interaction model $Y = (p(X)' \beta) T + f(Z) + e$, partialling out gives $\tilde{Y} = (p(X)' \beta) \tilde{T} + e$ with $\tilde{Y} = Y - E[Y | Z]$, $\tilde{T} = T - E[T | Z]$. So β is OLS of $\tilde{Y}$ on $p(X) \tilde{T}$, which is the R-learner loss of Section 3.5 with a linear CATE. The residuals come from `est_dml(model = "plr")`; the regression on $p(X) \tilde{T}$ is one `lm()` call with HC standard errors. The heterogeneity is then a slope that varies with X , exactly the “sensitivity” that price experiments estimate.

4.5 Forests with confidence intervals

Inference at a point x is harder than inference on a summary. Random forests adapted for inference (Wager and Athey 2018; Athey, Tibshirani, and Wager 2019) deliver it in low dimensions. The ingredients are:

- **honesty**: one subsample chooses the splits, another fills the leaves;
- **subsampling** without replacement, with subsample size $s \ll n$;
- **balanced splits** and a positive probability that each variable is tried at each split;
- fully grown trees with small leaves.

Under these conditions predictions are asymptotically normal, and a variance estimate (from the variation across subsamples) gives confidence intervals. Two constructions are in use.

- A **doubly robust forest** is an honest regression forest on the pseudo-outcome $Y(\hat{\eta})$. Its validity in high dimensions of Z needs only the product of nuisance errors to be small conditionally on X (Oprescu, Syrgkanis, and Wu 2019).
- A **causal forest** solves the R-learner moment locally: $E[(\tilde{Y} - \tau(x)\tilde{D})\tilde{D} \mid X = x] = 0$, using forest weights $K(x, X_i)$ that measure how often X_i shares a leaf with x . This is the generalized random forest of **grf**, and it is valid when the CATE depends on the forest's variables only.

The theory is exact only under the tuning it assumes; cross-validated hyperparameters break it. In practice the intervals are a useful signal of where the CATE is uncertain, not a guarantee. The comparison below uses the same data for `grf::causal_forest()` and for the DR-learner with a forest final stage.

```
library(grf)
cf <- causal_forest(as.matrix(train[, x5]), train$y, train$d, num.trees = 1000, seed = 1)
pred_cf <- predict(cf, as.matrix(test[, x5]), estimate.variance = TRUE)
dr_forest <- cate_learner(scores = sc_train_rf, x_het = x5, method = "dr",
  learner_final = lrn("regr.ranger", num.trees = 500, min.node.size = 50))
c(rmse_causal_forest = sqrt(mean((pred_cf$predictions - test$tau_true)^2)),
  rmse_dr_forest = sqrt(mean((predict(dr_forest, test) - test$tau_true)^2)),
  coverage_causal_forest = mean(abs(pred_cf$predictions - test$tau_true) <= 1.96 * sqrt(pred_cf$variance.estimate)))
#>      rmse_causal_forest      rmse_dr_forest coverage_causal_forest
#>           0.5195807           0.8014956           0.7880000
```

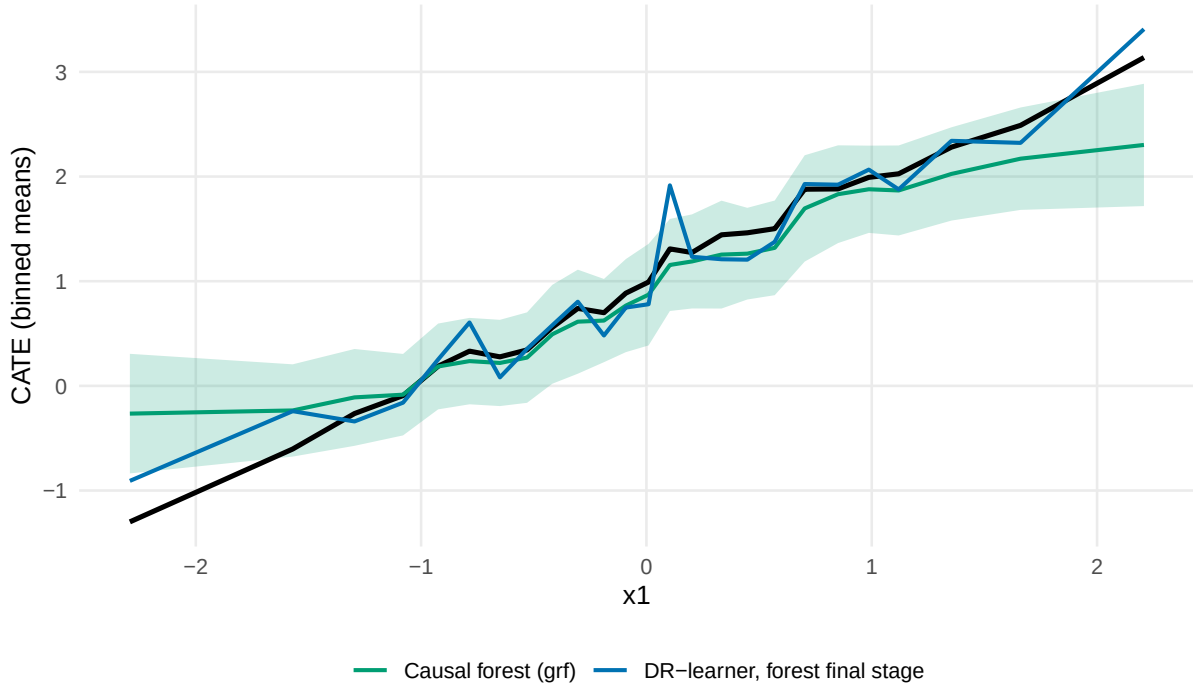


Figure 6: Causal forest (with pointwise 95 percent intervals) and DR-learner forest predictions along x_1 on the test sample, other covariates as observed. The black line is the true CATE averaged in bins of x_1 .

Both forests find the slope in x_1 ; the doubly robust one is noisier because it regresses a noisy label. The causal forest's pointwise intervals cover the true CATE for 79 percent of the test units, below the nominal 95 percent: the default tuning is not the tuning the theory assumes, and the intervals are a signal of uncertainty rather than a guarantee. Where the two forests disagree, the disagreement itself is informative: the R-learner moment weights regions by $\text{Var}(D \mid Z)$, the DR-learner does not (Section 3.5).

5 Choosing a CATE model out of sample

5.1 Why prediction metrics do not work

A CATE model cannot be scored by the usual out-of-sample mean squared error: the target $\tau(X_i)$ is never observed, and neither is $Y_i(1) - Y_i(0)$. Comparing models by how well they predict Y rewards the baseline, not the effect (Facure 2023, “Why prediction metrics are dangerous for causal models”). The pseudo-outcome solves this: it is an observed variable whose conditional mean is the target.

5.2 The doubly robust loss and its differences

For a candidate τ_m define, on a scoring sample,

$$\hat{L}(\tau_m) = \mathbb{E}_n[(Y(\hat{\eta}) - \tau_m(X))^2].$$

The loss itself contains the irreducible variance of the pseudo-outcome. Its differences do not: for two candidates,

$$L(\tau_i; \eta_0) - L(\tau_j; \eta_0) = E[(\tau_i(X) - \tau_0(X))^2] - E[(\tau_j(X) - \tau_0(X))^2].$$

Skeleton. Expand the squares; the cross term is $-2E[Y(\eta_0)(\tau_i - \tau_j)] = -2E[\tau_0(\tau_i - \tau_j)]$ by the key property, and $E[Y(\eta_0)^2]$ cancels in the difference. So ranking by $\hat{L}$ ranks by mean squared error against the true CATE. The difference is Neyman orthogonal in η , and Theorem 15.2.1 of Chernozhukov et al. (2026) gives $\sqrt{n}(\hat{L}_i - \hat{L}_j - \delta_{ij}^*) \rightarrow N(0, V)$ with V the variance of the per-observation loss difference (Appendix 10.3). Confidence intervals for the comparison follow.

The interpretable version compares each model with the constant-effect model: $S(\tau) = 1 - \hat{L}(\tau)/\hat{L}(\hat{\theta})$, in $(-\infty, 1]$, is the fraction of the constant model’s loss the candidate removes.

5.3 Code: scoring the meta-learners

We fit the five meta-learners on the training sample with forests, then score them on the scoring sample. The scoring sample gets its own cross-fitted pseudo-outcomes.

```
fit_one <- function(m, ...) cate_learner(scores = sc_train_rf, x = x5, x_het = c("x1", "x2"), method = m,
                                       learner = rf_first, learner_final = rf_final, ...)
models <- list(s_learner = fit_one("s"), t_learner = fit_one("t"), x_learner = fit_one("x"),
              dr_learner = fit_one("dr"), r_learner = fit_one("r"))
sc_score <- dr_scores(score_set, "y", "d", x5, learner_mu = rf_first, seed = 2)
cs <- do.call(cate_score, c(list(sc_score), models))
cs
#> Doubly robust loss of CATE models (n = 1000)
#>   constant model loss = 12.212; diff = baseline loss - model loss (positive is better)
#>   model   loss   diff std.error conf.low conf.high score_norm
#> s_learner 10.76  1.4476  0.1874  1.0804  1.815  0.118540
#> t_learner 10.61  1.6051  0.3087  1.0001  2.210  0.131439
#> x_learner 10.64  1.5736  0.2365  1.1101  2.037  0.128858
#> dr_learner 12.32 -0.1108  1.5336 -3.1167  2.895 -0.009072
#> r_learner 10.26  1.9473  0.5428  0.8835  3.011  0.159460
```

The doubly robust losses order the models `r_learner`, `t_learner`, `x_learner`, `s_learner`, `dr_learner`; the true root mean squared errors on the same rows, which only a simulation can compute, order them `x_learner`, `r_learner`, `t_learner`, `s_learner`, `dr_learner`. The confidence intervals say which differences are real: a candidate whose interval for `diff` includes zero has not been shown to beat a constant effect.

5.4 Ensembles

Selecting the single best model has a $\sqrt{\log(M)/n}$ excess risk. Stacking the candidates by minimizing the doubly robust loss over convex weights achieves M/n ; Q-aggregation, which adds the penalty $\sum_m w_m \hat{L}(\tau_m)$ to the stacked loss, achieves $\log(M)/n$, the best possible dependence on both M and n (Lecué and Rigollet

2014; Lan and Syrgkanis 2023). The intercept is fixed at the ATE and the base predictions are centered, so the ensemble never moves the average effect.

```
ens <- do.call(cate_ensemble, c(list(sc_score, method = "q"), models))
round(ens$ensemble$weights, 3)
#>  s_learner  t_learner  x_learner dr_learner  r_learner
#>  0.000      0.079      0.000      0.000      0.921
ens_rules <- sapply(c("q", "convex", "best", "ols"), function(r) {
  e <- do.call(cate_ensemble, c(list(sc_score, method = r), models))
  sqrt(mean((predict(e, test) - test$tau_true)^2))
})
best_single_rmse <- min(sapply(models, function(m) sqrt(mean((predict(m, test) - test$tau_true)^2))))
round(c(ens_rules, truth_best_single = best_single_rmse), 3)
#>               q          convex          best          ols
#>          0.594          0.505          0.635          0.620
#> truth_best_single
#>          0.469
```

The best base learner on the test sample has root mean squared error 0.469; the convex ensemble reaches 0.505, Q-aggregation 0.594, and the single best-scored model 0.635. No rule needs to know which base learner is best, and the convex rule comes closest. The scoring sample has only 1000 rows, so its ranking of the learners is imperfect (Section 5.3 showed the doubly robust losses and the true errors ordering the learners differently), and the ensembles inherit that noise. Unconstrained least squares can extrapolate outside the convex hull and is less stable.

5.5 What the pipeline cannot give

The ensemble comes without confidence intervals for its predictions and can change with the random split, the seeds of the forests, or a few removed observations. Report sensitivity to these choices. For statements with error control, hold out a third sample and use the tests of the next section.

6 Validating the chosen model

6.1 Three samples

Train on one sample, score and stack on a second, test on a third. The test sample sees the final model only as a fixed function τ^* . Everything in this section is computed on the test sample's own cross-fitted pseudo-outcomes.

```
sc_test <- dr_scores(test, "y", "d", x5, learner_mu = rf_first, seed = 3)
val <- cate_validate(sc_test, ens, n_boot = 999, seed = 1)
```

6.2 Heterogeneity test from the best linear predictor

Regress $Y(\hat{\eta})$ on $(1, \tau^*(X) - \bar{\tau}^*)$. The slope converges to

$$\beta_1 = \frac{\text{Cov}(\tau_0(X), \tau^*(X))}{\text{Var}(\tau^*(X))} = \frac{\text{Cov}(Y(1) - Y(0), \tau^*(X))}{\text{Var}(\tau^*(X))}.$$

A significant slope means the model is correlated with the individual effect, which also proves that heterogeneity in X exists. A slope of one means the model's scale is right. The intercept is the ATE because the regressor is centered.

```
val$blp
#>               term estimate std.error statistic      p.value   conf.low
#>               ATE 0.8217857 0.1111483   7.393594 1.429123e-13  0.6039389
#> tau_c tau_hat (centered) 0.5611208 0.3531791   1.588771 1.121121e-01 -0.1310975
#>               conf.high
#>               1.039632
#> tau_c 1.253339
```


Here the slope is 0.56 with standard error 0.35: the test does not reject a zero slope, while the grouped diagnostics below detect heterogeneity clearly. The slope is a regression on a heavy-tailed outcome and a few extreme predictions carry much of its variance; the grouped statistics average within quantile groups and are far more stable. Report both.

6.3 Calibration

A calibrated CATE model satisfies $E[Y(1) - Y(0) | \tau^*(X) = t] = t$. The mean squared error decomposes as calibration error plus distortion, $E[(\tau^* - \tau_0)^2] = \text{CAL}_2(\tau^*) + E[\text{Var}(\tau_0(X) | \tau^*(X))]$ (the proof is the variance decomposition with $E[\tau_0 | \tau^*]$ as the conditional mean), so any consistent model is eventually calibrated, but a model can be calibrated without being accurate. The binned check compares, in each quantile group of τ^* , the doubly robust GATE with the mean prediction.

```
val$calibration$table
#>   group    n share  mean_tau    gate std.error  conf.low  conf.high
#> 1     1   250  0.25 -0.6379013 -0.5251741 0.1895335 -0.89665301 -0.1536953
#> 2     2   250  0.25  0.3211336  0.2752399 0.1490372 -0.01686767  0.5673474
#> 3     3   250  0.25  1.1651649  1.1775116 0.1531030  0.87743532  1.4775879
#> 4     4   250  0.25  2.6615386  2.3595653 0.3289825  1.71477141  3.0043592
c(cal1 = val$calibration$cal1, cal1_normalized = val$calibration$cal1_normalized,
  p_equal = val$calibration$test_equal$p.value)
#>           cal1 cal1_normalized           p_equal
#> 1.182352e-01  8.751150e-01  7.135475e-18
```

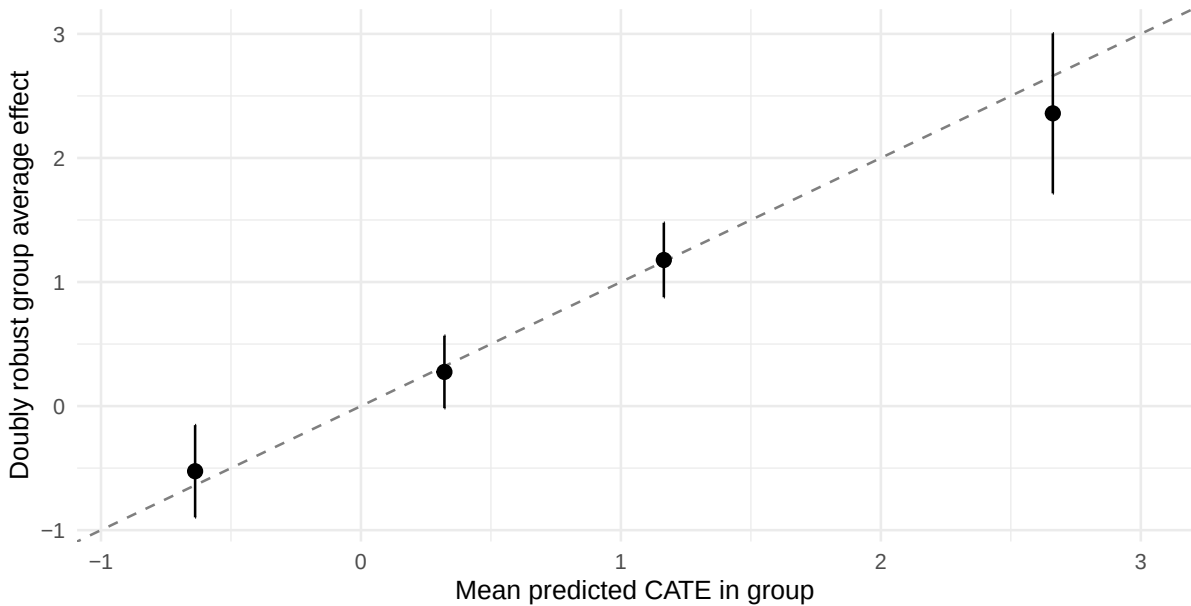


Figure 7: Calibration of the ensemble on the test sample: doubly robust group average effect (with 95 percent intervals) against the mean prediction in each quartile of predicted CATE. Points on the diagonal are calibrated.

6.4 Targeting curves

Suppose we can treat only a share q of the population and rank units by τ^* . The **TOC** curve (Chernozhukov et al. 2026, Section 15.3) is the average effect among the top q minus the ATE,

$$\text{TOC}(q) = E[Y(1) - Y(0) | \tau^*(X) \geq \mu(\tau^*, q)] - \text{ATE} = \text{Cov}\left(Y(1) - Y(0), \frac{\mathbb{1}\{\tau^*(X) \geq \mu(\tau^*, q)\}}{q}\right),$$

and the **QINI** curve scales it by the share treated, $\text{QINI}(q) = q \text{TOC}(q) = \text{Cov}(Y(1) - Y(0), \mathbb{1}\{\tau^*(X) \geq \mu(\tau^*, q)\})$. The covariance forms (Appendix 10.4) show that both are identified by replacing $Y(1) - Y(0)$

with $Y(\eta_0)$, so both are averages of the pseudo-outcome with fixed weights. Their influence functions give pointwise standard errors; the multiplier bootstrap gives a one-sided simultaneous band. If the band lies above zero at any q , the model has found heterogeneity with statistical significance. The areas under the curves, AUTOC and AUQC, integrate over all ranks and are the rank-weighted average treatment effects of Yadlowsky et al. (2021).

```
val$areas
#>   statistic estimate std.error conf.low    p.value
#> 1   AUTOC 0.8786673 0.37222647 0.2664092 9.123394e-03
#> 2   AUQC 0.3195273 0.04418044 0.2468569 2.373981e-13
val$heterogeneity_statistic
#>   curve max_lower_band at_q
#> 1   TOC      0.8965694 0.25
#> 2   QINI      0.3470092 0.55
```

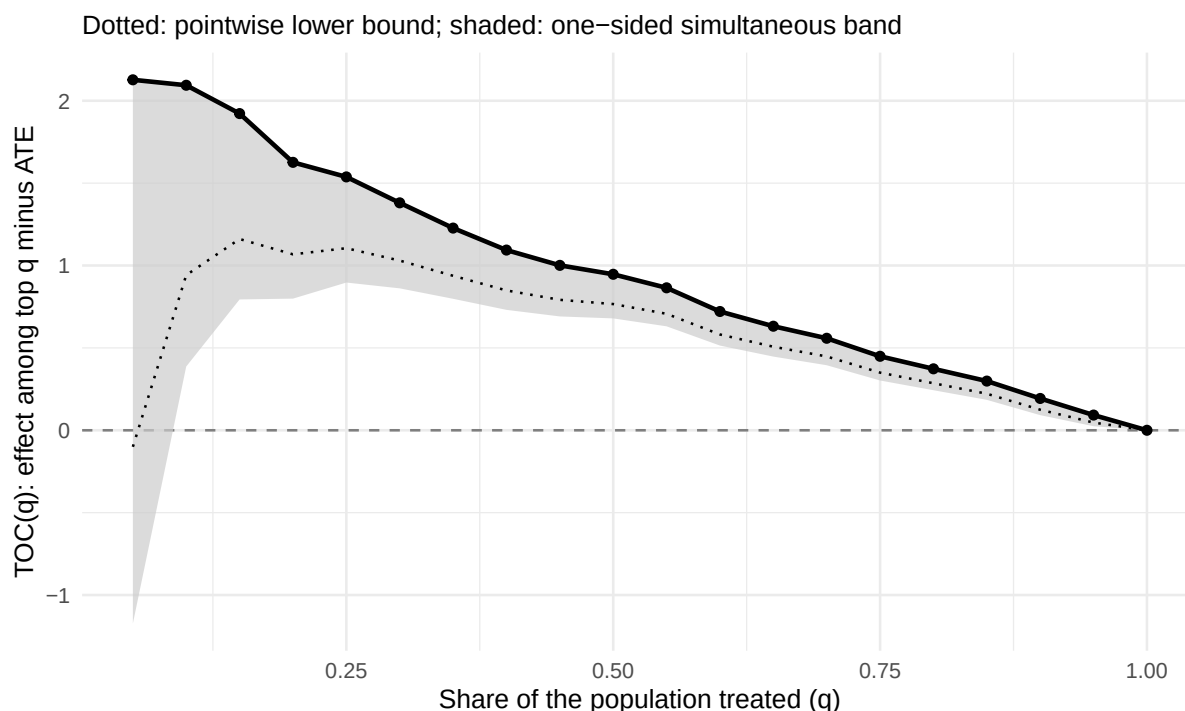


Figure 8: TOC curve of the ensemble on the test sample with pointwise (dotted) and one-sided simultaneous (shaded) 95 percent lower bounds. Treating the top 20 percent by predicted CATE yields an average effect about $\text{TOC}(0.2)$ above the ATE.

The TOC curve reads: treating the top 5 percent by predicted CATE yields an average effect 2.13 above the ATE of 0.82; the gain declines to zero as q reaches one. The QINI curve reads as total gain over treating a random share q ; it peaks at the share where marginal units stop being better than average.

6.5 Group differences

When the top and bottom groups are separated statistically, describing them is the last diagnostic: covariate means in the two groups, or a shallow classification tree that separates them.

```
val$group_diff %>% mutate(across(where(is.numeric), ~ round(.x, 3)))
#>   covariate mean_top se_top mean_bottom se_bottom diff se_diff
#> 1      x1      1.037 0.054      -1.089    0.043  2.126  0.069
#> 2      x2      0.090 0.090      -0.126    0.037  0.216  0.098
#> 3      x3     -0.053 0.067      -0.033    0.065 -0.021  0.093
#> 4      x4      0.047 0.065      -0.034    0.064  0.081  0.091
#> 5      x5     -0.088 0.065      -0.008    0.060 -0.080  0.089
```

In the design, x_1 drives the effect linearly and x_2 quadratically. The table shows the top quartile has a much higher mean of x_1 , while the mean of x_2 barely differs: a quadratic effect changes the spread of x_2 , not its mean, and a comparison of means cannot see it. x_3 to x_5 do not differ, as they should not.

6.6 Ties

A model that predicts the same value for many units (a tree, a coarse rule) leaves the top- q set ill defined. The package treats units above the threshold deterministically and units at the threshold with the probability that fills the share q (Chernozhukov et al. 2026, Remark 15.3.1); the areas use the average weight within each tied block, which is the expectation of that rule.

7 Policy learning

7.1 Policies and their value

A policy $\pi : X \rightarrow \{0, 1\}$ says whom to treat. Its gain over treating no one, net of a per-unit cost c , is

$$V(\pi) = E[\pi(X) (Y(1) - Y(0) - c)] = E[\pi(X) (Y(\eta_0) - c)],$$

the second equality by the key property of Section 2, since $\pi(X)$ is a function of X . So the value of a *fixed* policy is an average of the pseudo-outcome, with a standard error from its influence function. The gain over treating everyone is $E[(\pi(X) - 1)(Y(\eta_0) - c)]$. Under a budget that allows a share q to be treated, the best rule treats the top q by $\tau(X)$ and its value is $q \cdot (\text{TOC}(q) + \text{ATE})$ in the notation of Section 6.4.

```
policy_value(sc_test, list(sign_of_ensemble = predict(ens, test) > 0,
                    treat_all = rep(1, nrow(test)),
                    oracle = test$tau_true > 0), cost = 0.5)
#>           policy share_treated estimate std.error  conf.low conf.high
#> 1 sign_of_ensemble      0.722 0.5953772 0.1005322 0.3983377 0.7924168
#> 2   treat_all          1.000 0.3217857 0.1139273 0.0984923 0.5450790
#> 3    oracle           0.801 0.5501930 0.1028080 0.3486930 0.7516929
#>   diff_vs_first  diff_se
#> 1  0.00000000 0.00000000
#> 2 -0.27359157 0.05046374
#> 3 -0.04518428 0.02371220
```

With a cost of 0.5 per treated unit, treating everyone is clearly worse than the rule that treats units whose predicted effect exceeds the cost. The rule that knows the true CATE and the estimated rule are indistinguishable within sampling error: the difference between them is smaller than two of its standard errors.

7.2 Optimal policies are classifiers

Without constraints the optimal policy is $\pi^*(x) = \mathbb{1}\{\tau(x) > c\}$: only the *sign* of $\tau(x) - c$ matters. Learning a policy is therefore closer to classification than to regression. Mistakes cost more where $|\tau(x) - c|$ is large, so it is cost-sensitive classification. Formally, with $\Gamma = Y(\eta_0) - c$,

$$\arg \max_{\pi} V(\pi) = \arg \max_{\pi} E[(2\pi(X) - 1)\Gamma] = \arg \max_{\pi} E[\mathbb{1}\{2\pi(X) - 1 = \text{sign}(\Gamma)\} |\Gamma|].$$

Skeleton. $(2\pi - 1)\Gamma = (2\pi - 1)\text{sign}(\Gamma)|\Gamma|$, and $(2\pi - 1)\text{sign}(\Gamma)$ equals $+1$ when the policy matches the sign and -1 otherwise; adding the constant $E|\Gamma|$ and halving gives the indicator form. So any weighted classifier, trained with labels $\text{sign}(\hat{\Gamma}_i)$ and weights $|\hat{\Gamma}_i|$, maximizes the empirical value.

7.3 Empirical welfare maximization in a restricted class

Plugging a CATE model into the sign rule is one option. Empirical welfare maximization (Kitagawa and Tetenov 2018; Athey and Wager 2021) is another: choose the rule in a class Π that maximizes the sample

value $\hat{V}(\pi) = \mathbb{E}_n[\pi(X)\hat{\Gamma}]$. Restricting Π has two purposes. A depth-two tree or a linear rule can be explained, audited, and implemented, which a forest cannot. And the regret of the learned rule,

$$R(\hat{\pi}) = \max_{\pi \in \Pi} V(\pi) - V(\hat{\pi}) \lesssim \sqrt{\frac{V^* \text{VC}(\Pi)}{n}} + \|H(\hat{p}) - H(p_0)\| \|\hat{g} - g_0\|,$$

is controlled by the complexity of Π (its VC dimension), not by the complexity of the CATE (Athey and Wager 2021, Theorem 1; Appendix 10.5 sketches the reduction). The nuisance errors enter as a product, as everywhere else in this lecture. The plug-in rule inherits the CATE model's estimation error instead, which is larger when the CATE is harder to learn than its sign.

7.4 Code: learning a tree

The "policy" design has a CATE that is piecewise constant on a depth-two tree in x_1, x_2, x_3 (plus a small smooth term in x_4), confounded assignment, and a nonlinear baseline. `policy_learn()` searches depth-one and depth-two trees exactly, holding out half of the rows to report an honest value.

```
pol_dat <- sim_hte(4000, dgp = "policy", seed = 2)
sc_pol <- dr_scores(pol_dat, "y", "d", x5, learner_mu = rf_first, seed = 1)
pol <- policy_learn(sc_pol, method = "tree", depth = 2, seed = 1)
pol
#> Treatment policy (tree, depth 2, exhaustive)
#> cost = 0, share treated = 0.382
#> if x1 <= 0.02754:
#>   if x3 <= -0.2485:
#>     action = 1 (treat)
#>   else:
#>     action = 0 (do not treat)
#> else:
#>   if x2 <= 0.2487:
#>     action = 0 (do not treat)
#>   else:
#>     action = 1 (treat)
#> value over no treatment / over treating everyone:
#>   sample      n share_treated value_vs_none se_vs_none value_vs_all se_vs_all
#> estimation 2000      0.3675      0.4992      0.03281      0.5058      0.03918
#> holdout 2000      0.3975      0.5045      0.03486      0.4567      0.03711
attr(pol_dat, "optimal_tree")
#> $root
#> [1] "x1 <= 0"
#>
#> $left
#> [1] "x3 <= -0.25: treat (tau = 1.5 + 0.3 x4); else do not treat (tau = -0.5 + 0.3 x4)"
#>
#> $right
#> [1] "x2 <= 0.25: do not treat (tau = -1 + 0.3 x4); else treat (tau = 1 + 0.3 x4)"
```

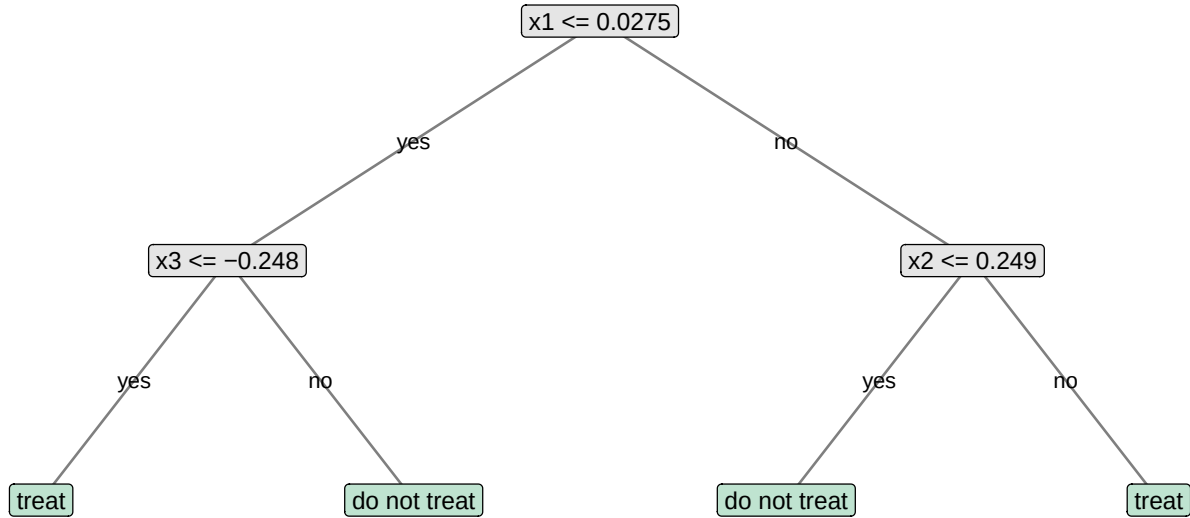


Figure 9: The depth-two policy tree learned by empirical welfare maximization on the policy design. It recovers the structure of the optimal rule.

The estimation-sample value is optimistic: it is the maximum over many candidate trees. The held-out value is not. `engine = "policytree"` delegates the search to the `policytree` package (Sverdrup et al. 2020), which handles any depth with the same objective:

```

pol_pt <- policy_learn(sc_pol, method = "tree", depth = 2, engine = "policytree", seed = 1)
c(agreement = mean(predict(pol_pt, pol_dat) == predict(pol, pol_dat)))
#> agreement
#> 0.999

```

7.5 Regret of trees versus plug-in rules

Figure 10 compares four rules on the policy design as n grows: depth-one and depth-two trees from empirical welfare maximization, and the signs of T- and DR-learner CATE models. Regret is measured with the true CATE on a large test sample.

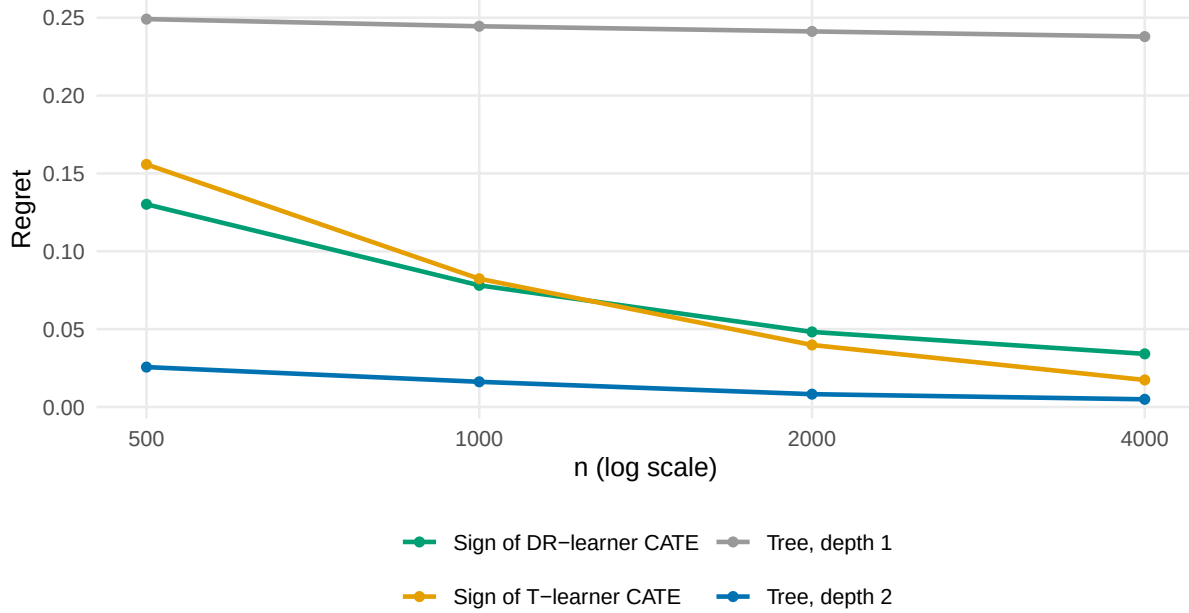


Figure 10: Regret (value of the optimal rule minus value of the learned rule, in outcome units) on the policy design, mean over simulations with a two-standard-error band. The depth-two tree has the right class and learns fastest; the depth-one tree has the wrong class and its regret does not vanish; plug-in sign rules need to learn the magnitude before the sign.

Two lessons. The class must contain a good rule: the depth-one tree cannot express the optimal policy and its regret stays flat. Within the right class, learning the sign directly is easier than learning the magnitude: at every n the depth-two tree beats the plug-in rules built from forests.

7.6 Budgets and other classes

- **Budget:** `policy_learn(method = "budget", budget = q, tau_hat = model)` treats the top share q ranked by a CATE model, the constrained optimum when the ranking is right. Its value is the QINI curve at q plus q times the ATE.
- **Linear rules** $\mathbb{1}\{x'b > 0\}$: Kitagawa and Tetenov solve the exact problem by integer programming; the package fits the weighted logistic surrogate, which is fast and usually close.
- **Any classifier:** `method = "classifier"` passes labels and weights to an `mlr3` classifier; deeper trees or forests give flexible but opaque rules.

```
tau_pol <- cate_learner(scores = sc_pol, x_het = x5, method = "dr", learner_final = rf_final)
rules <- list(
  tree2 = pol,
  budget30 = policy_learn(sc_pol, method = "budget", budget = 0.3, tau_hat = tau_pol, seed = 1),
  linear = policy_learn(sc_pol, method = "linear", seed = 1)
)
do.call(rbind, lapply(names(rules), function(nm) cbind(rule = nm, rules[[nm]]$value[2, ])))
```

#>	rule	sample	n	share_treated	value_vs_none	se_vs_none	value_vs_all
#> 2	tree2	holdout	2000	0.3975	0.5045069	0.03486118	0.4567212
#> 21	budget30	holdout	2000	0.3215	0.5715737	0.03152566	0.5237880
#> 22	linear	holdout	2000	0.4910	0.3051814	0.03772836	0.2573958

```
#> se_vs_all
#> 2 0.03710523
#> 21 0.03910547
#> 22 0.03634030
```

7.7 Targeting impact versus deprivation

Haushofer, Niehaus, Paramo, Miguel, and Walker (2025) ask whom a cash transfer should reach: the poorest, or those who gain most. With a concave utility u and predicted untreated outcome $\hat{y}_0$ (deprivation) and effect $\hat{\tau}$ (impact), a planner treats units in order of $u(\hat{y}_0 + \hat{\tau}) - u(\hat{y}_0)$. Linear utility ranks on impact; strong curvature ranks on deprivation. `policy_frontier()` traces the overlap of the chosen set with impact-only and deprivation-only targeting as curvature grows, and evaluates each rule with the doubly robust value.

```
fr <- policy_frontier(sc_test, ens, y0_hat = sc_test$nuisance$mu0, budget = 0.3,
                     utility = "cara", curvature = c(0, 0.5, 1, 2, 5))

fr
#> Targeting frontier (cara utility, budget = 0.3, n = 1000)
#> curvature overlap_impact overlap_deprivation mean_tau_selected
#>      0.0      1.0000      0.00000      2.4619
#>      0.5      0.6067      0.04333      1.8114
#>      1.0      0.3033      0.15333      1.1258
#>      2.0      0.1900      0.24000      0.8515
#>      5.0      0.1433      0.26000      0.7385
#> mean_y0_selected value std.error
#>      2.1934 0.6607 0.09042
#>      1.3280 0.4955 0.07286
#>      0.8345 0.3156 0.04512
#>      0.5995 0.2505 0.04210
#>      0.5499 0.2137 0.04310
```

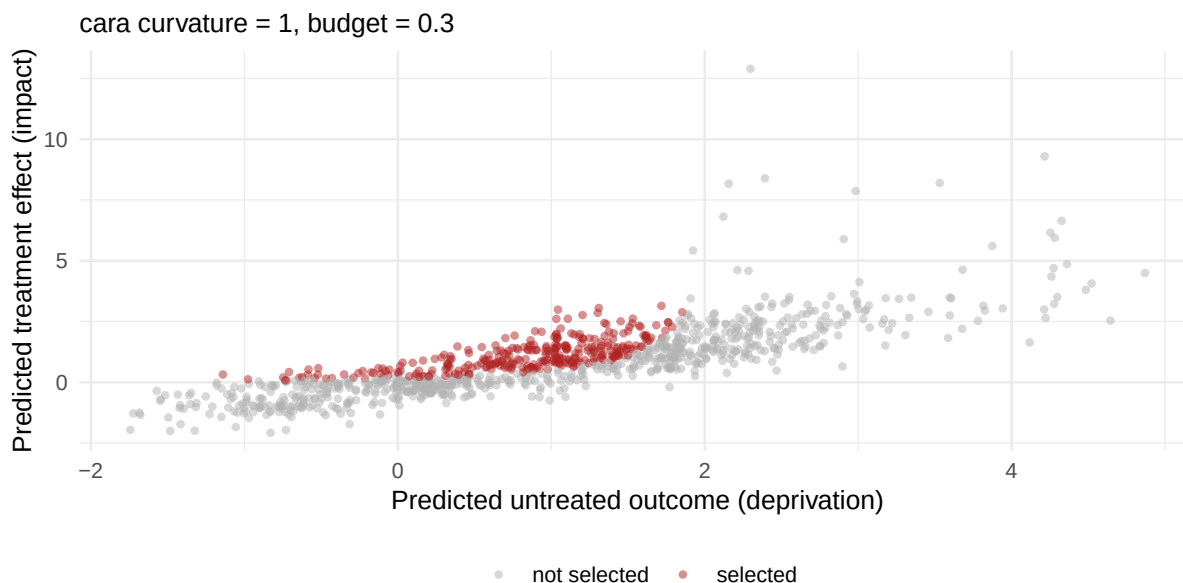


Figure 11: Units selected by a planner with CARA curvature 1 and a 30 percent budget, in the plane of predicted untreated outcome and predicted effect. The planner trades impact for deprivation along the frontier.

In the running design impact and deprivation are only weakly related, so the planner's choice moves smoothly from one corner to the other; the value of the rule falls as curvature rises because the value counts effects, not utility. The paper's finding is that in Kenya the two criteria pick different households, so deprivation-only targeting leaves impact on the table.

7.8 Cautions

- **Variance and overlap.** A candidate policy that treats where the data rarely treated has a noisy value; regret bounds carry the worst overlap ratio. Variance penalization, distributionally robust objectives, and pessimistic (lower-confidence-bound) objectives address this, at the cost of leaving the classification

reformulation.

- **Adaptive data.** When assignment probabilities were updated during collection (bandits), the pseudo-outcome must use the realized probabilities and stabilizing weights (Zhan, Ren, Athey, and Zhou 2024).
- **Dynamic regimes.** Sequential treatments need backward induction over stages; Section 07’s L19 material and Q-learning cover it.
- **Honesty.** Report the value on data the search never saw. The in-sample value of the best of many rules is biased upward.

8 Traps

8.1 Binary outcomes and the curvature trap

With a binary outcome the effect on the probability scale depends on where a unit sits on the response curve. Facure (2023, ch. 23) gives the cleanest example. A randomized nudge, income that moves only the baseline, and age that moves only the latent effect:

$$Y^* = -4.5 + 0.001 \text{ income} + D(1 + 0.01 \text{ age}) + \varepsilon, \quad Y = \mathbb{1}\{Y^* > t\}.$$

On the latent scale, income has no effect heterogeneity. On the probability scale it does, through the slope of the normal CDF, and it can dominate age.

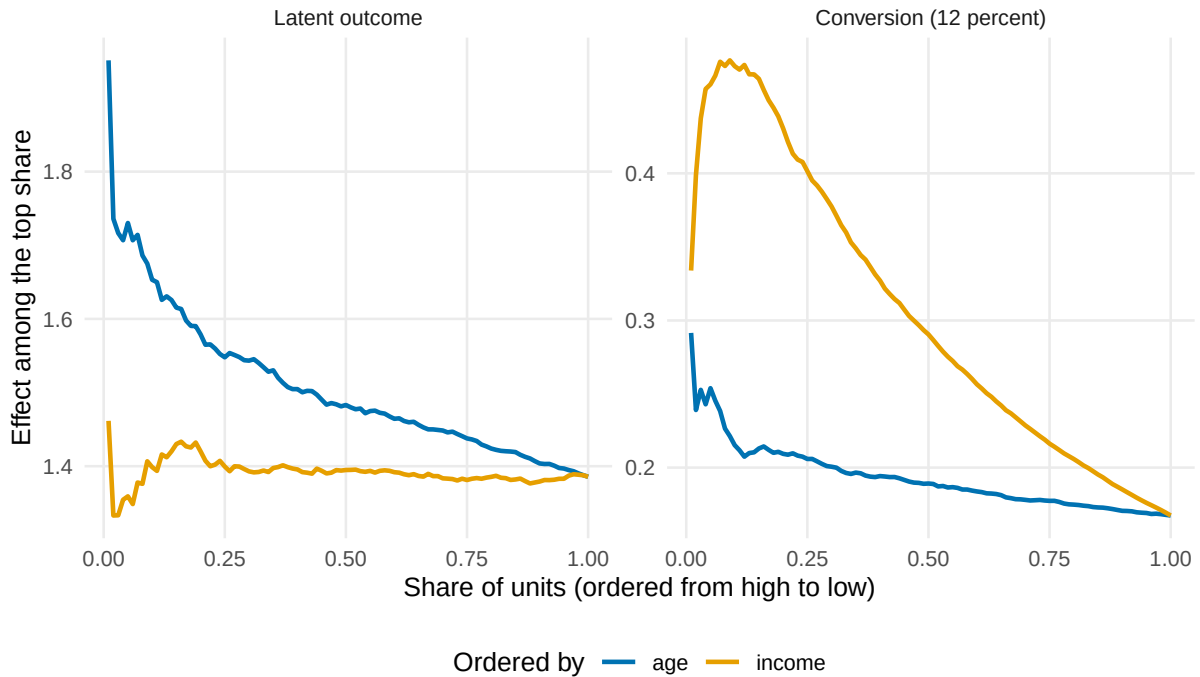


Figure 12: Cumulative effect curves (Facure 2023): units ordered by a covariate, difference in means between treated and controls among the top share. Left: latent outcome, where only age matters. Right: the binary conversion with a 12 percent base rate, where income (which moves only the baseline) sorts units by treatment effect more sharply than age.

The trap has a use. When conversion is rare, units with a high baseline are on the steep part of the curve and respond most; when conversion is common, units with a low baseline do. A CATE model on the probability scale is right about the effect on conversion, which is what a business cares about, but the ranking it produces is driven by the baseline, not by the latent sensitivity. Decide which scale the decision lives on before modelling heterogeneity.

8.2 Extrapolation of parametric summaries

A best linear predictor in a polynomial of one covariate is fitted where the data are dense and extrapolated where they are not. Heavy tails in the covariate produce wild curves in the tails, while forests regularize toward the nearest data (Chernozhukov et al. 2026, Section 15.5). Plot the covariate’s distribution next to the curve, and report the curve only over the central range.

8.3 Specification search

CATE estimates are noisy and rarely the paper’s main object, so they are prone to being reported when interesting and dropped when not (Childers 2024). Pre-register the heterogeneity dimensions or the analysis pipeline; use held-out data for every claim; treat data-driven subgroups as hypotheses to be tested in the next experiment.

8.4 Heterogeneity that is not the CATE

Two other questions hide under “heterogeneous effects”.

- **Effects on distributions.** Quantile treatment effects $Q_{Y(1)}(u) - Q_{Y(0)}(u)$ and effects on the CDF are identified by the same adjustment formulas as the mean, applied to $\mathbb{1}\{Y \leq y\}$ (distribution regression) or to the check loss (quantile regression; Chernozhukov, Fernández-Val, and Melly 2013). Parallel trends is the one design where this fails, because the assumption is about means (Athey and Imbens 2006).
- **Distributions of effects.** The distribution of $Y(1) - Y(0)$ is *not* identified without further assumptions, because the two potential outcomes are never observed together. Only bounds (Makarov bounds from the Fréchet-Hoeffding inequalities) or functional-form assumptions deliver it.

The CATE is neither: it is the conditional mean of the effect, the one feature of the effect distribution that adjustment identifies at every x .

```
qte <- sapply(c(0.1, 0.25, 0.5, 0.75, 0.9), function(u) {
  q1 <- quantreg::rq(y ~ 1, tau = u, data = subset(bin, d == 1))
  q0 <- quantreg::rq(y ~ 1, tau = u, data = subset(bin, d == 0))
  coef(q1) - coef(q0)
})
round(setNames(qte, paste0("QTE at ", c(0.1, 0.25, 0.5, 0.75, 0.9))), 3)
#> QTE at 0.1 QTE at 0.25 QTE at 0.5 QTE at 0.75 QTE at 0.9
#> 0 0 0 0 1
```

Quantile effects of the nudge on the binary outcome are zero at every quantile except where the treated distribution crosses the threshold; on the latent outcome they would all be about 1.4. The two scales tell different stories, again.

9 Practice

9.1 Replications and anchors

Four papers anchor this section; their replication notes live next to the lecture in the notes repository.

- **Athey and Wager (2021), policy learning with observational data.** Depth-limited policy trees from doubly robust scores in the California GAIN experiments; the simulation evidence (their Figure 2 and Appendix Table III) is reproducible from the saved outputs, and the tree search is `policy_learn()` or `policytree`.
- **Haushofer, Niehaus, Paramo, Miguel, and Walker (2025), targeting impact versus deprivation.** Causal forests predict untreated consumption and the transfer effect; a planner with CARA or CRRA utility trades them off. `policy_frontier()` reproduces the exercise from any pair of predictions.

- **Yoganarasimhan, Barzegary, and Pani (2023), optimal free trials.** Seven trial lengths as arms, lasso, boosting, causal trees and forests compared by out-of-sample scoring; a multi-arm case that reduces to pairwise contrasts and the scoring of Section 5.
- **Zhan, Ren, Athey, and Zhou (2024), policy learning with adaptively collected data.** Doubly robust scores with adaptive weights when the assignment probabilities changed during the experiment.

9.2 A checklist for heterogeneity and targeting

1. State the question: description (for whom), or decision (whom to treat). The second needs the value and a policy class, not the CATE's magnitude.
2. Split the data: train, score, test. Report which sample produced each number.
3. Build the pseudo-outcome with cross-fitted nuisances. Report the nuisance learners, their fit, and the propensity distribution.
4. For inference on summaries, use the best linear predictor or GATEs with simultaneous bands. Say which covariates define heterogeneity and which handle confounding.
5. For flexible models, fit several meta-learners and forests, score them on the scoring sample with confidence intervals, and stack.
6. Validate the chosen model on the test sample: heterogeneity slope, calibration, TOC and QINI with bands, AUTOC.
7. For policies, learn in a class you can implement; report the held-out value with a standard error, against treating everyone and no one.
8. Check stability across seeds and splits; state the scale (latent, probability, money) the decision lives on.

10 Appendix

10.1 Conditional orthogonality of the pseudo-outcome

Let $\eta = (p, g)$ and write $\mu_d = g(d, \cdot)$. Conditioning on Z ,

$$E[Y(\eta) \mid Z] = \mu_1 - \mu_0 + \frac{p_0}{p} (\mu_{1,0} - \mu_1) - \frac{1-p_0}{1-p} (\mu_{0,0} - \mu_0),$$

where $\mu_{d,0} = g_0(d, Z)$. Collect terms:

$$E[Y(\eta_0) - Y(\eta) \mid Z] = \left(1 - \frac{p_0}{p}\right) (\mu_{1,0} - \mu_1) - \left(1 - \frac{1-p_0}{1-p}\right) (\mu_{0,0} - \mu_0) = (H(p_0) - H(p)) (g_0(D, Z) - g(D, Z))$$

in conditional expectation, since $E[D \mid Z] = p_0$. Each factor vanishes at η_0 , so the Gateaux derivative in any direction is zero, and the bias is the product of the two errors. Averaging over Z given X gives the statement of Section 2.

10.2 Error bounds of the DR- and R-learners

DR-learner. Let $\hat{\tau}$ be the oracle's output on $\{X_i, Y_i(\hat{\eta})\}$ and $\bar{\tau}$ its output on the true labels $\{X_i, Y_i(\eta_0)\}$. For square-loss oracles that are stable (a Lipschitz map from labels to predictions, as least squares in a fixed class or a regularized learner under standard conditions), $\|\hat{\tau} - \bar{\tau}\|_{L_2(X)} \lesssim \|E[Y(\hat{\eta}) - Y(\eta_0) \mid X]\|_{L_2(X)} + (\text{noise term})$. The first term equals $\|E[(H(p_0) - H(\hat{p}))(g_0 - \hat{g}) \mid X]\|$ by Appendix 10.1, bounded by the product of the nuisance L_4 errors by Cauchy-Schwarz. The noise term is second order because the label perturbation has conditional mean given by the product and conditional variance that vanishes with consistency. Then $\|\hat{\tau} - \tau^*\| \leq \|\bar{\tau} - \tau^*\| + \|\hat{\tau} - \bar{\tau}\|$ gives $r_n + \text{Err}(\hat{g})\text{Err}(H(\hat{p}))$; squaring gives the bound in Section 3.4. Foster and Syrgkanis (2023) give the general theorem (orthogonal statistical learning), Kennedy (2023) sharp rates for specific oracles.

R-learner. The population loss is $L(\tau; \eta) = E[(Y - l(Z) - \tau(X)(D - p(Z)))^2]$. Its derivative in τ at (τ_0, η_0) is orthogonal to η in the sense that $\partial_{\eta} \partial_{\tau} L(\tau_0; \eta_0) = 0$, which is the partialling-out orthogonality of

Section 04 applied conditionally. The second-order remainder contains $E[(\hat{p} - p_0)^2(\tau - \tau_0)^2]$ -type terms and $E[(\hat{p} - p_0)(\hat{l} - l_0)(\tau - \tau_0)]$; bounding them by Cauchy-Schwarz gives the $\text{Err}(\hat{p})^4$ and $\text{Err}(\hat{p})^2 \text{Err}(\hat{l})^2$ terms of Section 3.5 (Nie and Wager 2021, Theorem 1; Foster and Syrgkanis 2023).

10.3 Loss differences are asymptotically normal

Let $\Delta_i = (Y_i(\eta) - \tau_i)^2 - (Y_i(\eta) - \tau_j)^2 = (\tau_i - \tau_j)(\tau_i + \tau_j - 2Y(\eta))$ evaluated at X_i . Then $\hat{\delta}_{ij} = \mathbb{E}_n[\Delta_i]$ is a sample mean whose population value at η_0 is $E[(\tau_i - \tau_0)^2] - E[(\tau_j - \tau_0)^2]$ (expand and use $E[Y(\eta_0) | X] = \tau_0$). The dependence on η enters only through $E[Y(\eta) | X]$, so the functional is Neyman orthogonal and its nuisance error is the product bound of Appendix 10.1 weighted by $\tau_i - \tau_j$. Under $\sqrt{n}\|H(\hat{p}) - H(p_0)\|_{L_2}\|\hat{g} - g_0\|_{L_2} \rightarrow 0$ and consistency, $\sqrt{n}(\hat{\delta}_{ij} - \delta_{ij}^*) \rightarrow N(0, V)$ with $V = \text{Var}(\Delta)$, estimated by the sample variance of $\hat{\Delta}_i$ (Chernozhukov et al. 2026, Theorem 15.2.1). The variance is bounded below by $4E[(\tau_i - \tau_j)^2 \text{Var}(Y(\eta_0) | X)]$ (their Appendix 15.A), which is why models must be separated by more than the estimation error for the comparison to be informative.

10.4 TOC and QINI as covariances

Let $A = Y(1) - Y(0)$ and $B_q = \mathbb{1}\{\tau^*(X) \geq \mu(\tau^*, q)\} / P(\tau^*(X) \geq \mu(\tau^*, q))$, so $E[B_q] = 1$. Then $\text{TOC}(q) = E[A | B_q > 0] - E[A] = E[AB_q] - E[A]E[B_q] = \text{Cov}(A, B_q)$. Likewise $\text{QINI}(q) = (E[A | B_q > 0] - E[A])P(B_q > 0) = \text{Cov}(A, \mathbb{1}\{B_q > 0\})$. Since B_q is a function of X and $E[A | X] = E[Y(\eta_0) | X]$, both covariances are unchanged when A is replaced by $Y(\eta_0)$. The influence function of $\widehat{\text{TOC}}(q)$ is $\psi_q = (Y(\eta_0) - \theta_0)(\mathbb{1}\{\cdot\} / \pi_0(q) - 1) - \text{TOC}(q)$ with θ_0 the ATE and $\pi_0(q)$ the treated share; the estimated ATE and share are nuisances whose first-order effect vanishes because the functional is orthogonal to both (the covariance form is invariant to the centering). Joint normality over a grid of q and the simultaneous band follow from the high-dimensional central limit theorem and the multiplier bootstrap of Section 07 (Chernozhukov et al. 2026, Theorem 15.3.1). Integrating over all q with $u = k/n$ gives the rank weights of Section 6.4: the unit ranked r contributes $\sum_{k \geq r} 1/k - 1$ to AUTOC.

10.5 Regret of empirical welfare maximization

Let $\hat{\pi} = \arg \max_{\pi \in \Pi} \hat{V}(\pi)$ with $\hat{V}(\pi) = \mathbb{E}_n[\pi(X)\hat{\Gamma}]$ and $\pi^* = \arg \max_{\pi \in \Pi} V(\pi)$. Then

$$V(\pi^*) - V(\hat{\pi}) = [V(\pi^*) - \hat{V}(\pi^*)] + [\hat{V}(\pi^*) - \hat{V}(\hat{\pi})] + [\hat{V}(\hat{\pi}) - V(\hat{\pi})] \leq 2 \sup_{\pi \in \Pi} |\hat{V}(\pi) - V(\pi)|,$$

because the middle bracket is at most zero by the definition of $\hat{\pi}$. The supremum splits into the oracle part, with the true Γ , and the nuisance part. The oracle part is the supremum of an empirical process indexed by Π and is $O_p(\sqrt{V^* \text{VC}(\Pi)/n})$ by a uniform law of large numbers with the variance proxy V^* . The nuisance part is $\sup_{\pi} |\mathbb{E}_n[\pi(X)(\hat{\Gamma} - \Gamma)]|$, whose leading term is the product of nuisance errors by the orthogonality of the pseudo-outcome, with cross-fitting removing the dependence between $\hat{\eta}$ and the sample (Athey and Wager 2021, Theorem 1 and Lemma 2).

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